

CHILE–HONDURAS MEETING

Differential Equations in Biological Sciences and Engineering

University of Talca

Faculty of Agricultural Sciences

BOOK OF ABSTRACTS

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Research and Transfer Center for Irrigation and Agroclimatology (CITRA),

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Introduction

The history behind this meeting is deeply rooted in academic collaboration, friendship, and a shared passion for mathematics across Latin America.

The origins of this initiative can be traced back to Dr. Marlon López-Flores, a Honduran mathematician whose academic career led him from Honduras to advanced studies in Brazil. In 2010, during a Summer School at the Instituto de Matemática Pura e Aplicada (IMPA), Dr. López-Flores met the Chilean mathematician Dr. William Campillay-Llanos. During that same event, they also met a young mathematics enthusiast, José Ramón Madrid Padilla, who would later become an internationally recognized researcher. Although their careers would develop along different paths, the scientific ties established during those formative years remained active.

Fifteen years later, in 2025, these researchers met again during the 35th Brazilian Mathematics Colloquium. Motivated by common interests in mathematical modeling, differential equations, dynamical systems, and interdisciplinary applications, they began discussing the possibility of establishing a long-term scientific collaboration between Chile and Honduras. These conversations eventually led to the creation of a broader academic initiative involving researchers, graduate students, and institutions from both countries.

A decisive step in this process was the formation of the **Grupo de Biomodelamiento Matemático e Inteligencia Artificial (GBM-IA)**, a collaborative research group composed primarily of young Honduran researchers from the Master's Program in Mathematics at the National Autonomous University of Honduras (UNAH), together with international collaborators. The group was conceived with the goal of developing advanced mathematical methodologies for problems arising in biological sciences, engineering, artificial intelligence, and data-driven modeling.

The scientific foundations of the GBM-IA are supported by an active research agenda in applied analysis, differential equations, numerical methods, and mathematical modeling. This collaboration has already produced several ongoing manuscripts that develop new mathematical frameworks for positive-state systems, proportional arithmetic, logarithmic discretization, and computational modeling.

Among the theoretical contributions are the manuscripts *A Proportional-Arithmetic Framework for Fourier Analysis on the Positive Real Line*, *A Positive-State Laplace Calculus for Dirichlet Resolvents and Log-Ratio Stability*, and *Logarithmic-Scale Finite Difference Schemes for Positive Parabolic Problems on Geometric Grids*. These studies establish novel analytical and numerical methodologies for the treatment of positive dynamical systems and provide the mathematical foundations for future applications in biological sciences, engineering, and artificial intelligence.

Building upon these theoretical developments, the collaboration has expanded toward data-driven mathematical modeling through the conference contribution *Proportional Discrete Fourier Analysis of WaPOR Actual Evapotranspiration over Cropland in Comayagua, Honduras*, together with the ongoing studies *A Proportional Positive-State Model for Post-Irrigation Soil-Moisture Recovery Using Open Smart-Irrigation Data* and *Multiyear Validation and Mass-Balance Auditing of Richards Simulations on Uniform and Graded Finite-Volume Meshes Using USDA SCAN Observations*. These projects combine mathematical modeling, numerical analysis, remote sensing, artificial intelligence, and environmental data to address contemporary problems in agriculture and water-resource management.

Together, these works illustrate the scientific vision of the GBM–IA: developing rigorous mathematical theory while simultaneously creating computational methodologies capable of addressing real-world challenges in biological sciences, engineering, agriculture, and artificial intelligence.

The initiative was significantly strengthened through the scientific leadership of Dr. Samuel Ortega-Farías, Director of the Research and Transfer Center for Irrigation and Agroclimatology (CITRA) at the University of Talca. Over the past decade, Dr. Ortega-Farías has been one of the principal promoters of biomathematical modeling in agricultural sciences, fostering the integration of mathematical modeling, agroclimatology, crop physiology, environmental physics, and data-driven methodologies. His interdisciplinary vision has been reflected not only in his scientific contributions but also in the development of postgraduate education, where he has actively promoted advanced training in mathematical modeling and its applications to agriculture and environmental sciences.

This vision also inspired the collaboration between Dr. Marlon López-Flores and Dr. William Campillay-Llanos at CITRA through the international academic project *Projection of Sustainable Fruit Production through Biomathematical Modeling and Artificial Intelligence* (International Academic Events Fund, 2023). This project strengthened scientific cooperation between Chile and Honduras, bringing together expertise in mathematical modeling, artificial intelligence, agroclimatology, and agricultural sciences, ultimately laying the foundations for the present international meeting.

More recently, the integration of advanced mathematical modeling, stochastic processes, and agricultural phenology has opened new interdisciplinary research directions. A representative example is the work

Rosales, A., Ortega-Farías, S., de la Fuente-Sáiz, D., Morales-Zárate, R., Campillay-Llanos, W., Suarez, E., & Lisperguer, M. J. (2026). *Development of Predictive Models for Simulating Vegetative and Reproductive Phenology in European Hazelnut Trees (Corylus avellana L.)*. International Journal of Biometeorology, 70(6), 184.

which illustrates how mathematical and stochastic modeling can improve the understanding of plant development under changing environmental conditions, providing robust tools for precision agriculture, climate adaptation, and sustainable crop management.

To further consolidate this collaborative effort, Dr. Braulio Soto, Director of the Núcleo de Investigación en Producción Alimentaria (NIPA) and Editor-in-Chief of the *Journal of Natural Resources and Farming*, joined the scientific and organizing committee. His participation strengthened the institutional framework of the initiative and promoted the creation of the Special Issue entitled *Mathematical and Computational Modelling of Agricultural and Natural Resource Systems*, providing an international publication venue aligned with the scientific objectives of the meeting.

This collaboration culminated in the organization of the

Chile–Honduras Meeting

Differential Equations in Biological Sciences and Engineering

María Clara Fittipaldi

Title

Asymptotic behavior of a logistic epidemiological model with random transmission and treatment rates

Affiliation:

Faculty of Sciences, National Autonomous University of Mexico (UNAM), Mexico

Abstract

In this talk, we present an extension of the stochastic logistic treatment model for AIDS transmission and control introduced by Ding, Xu, and Hu (2009), itself a modification of the classical Roberts and Saha (1999) model, by incorporating a second independent Brownian noise source.

Unlike the Ding-Xu-Hu model, whose dynamics are perturbed by a single environmental noise source, our formulation includes two distinct stochastic perturbations acting simultaneously on the proportion of infected individuals. This leads to a stochastic differential equation with logistic drift and a diffusion term whose intensity depends on the infected proportion.

We show that the model admits an equivalent-in-law representation driven by a single Brownian motion constructed through an adapted combination of the original noises. Using this equivalent representation, we derive a stochastic reproduction number separating extinction and persistence regimes, and compare our persistence threshold with those obtained in the Roberts-Saha and Ding-Xu-Hu models.

This is joint work with Alejandra Christen (University of Valparaíso, Chile) and José Miguel Zúñiga Núñez (Catholic University of Maule, Chile).

Keywords:

Stochastic differential equations, epidemiological models, persistence, extinction, AIDS dynamics.

Ángela Carolina Tunubalá Sánchez

Title

Limit Cycles on Hybrid Systems

Affiliation:

Institute of Mathematics and Statistics, Federal University of Goiás, Goiânia, Goiás, Brazil

Department of Mathematics, Autonomous University of Barcelona, Barcelona, Spain

Abstract

A hybrid dynamical system is one whose behavior is governed by both continuous and discrete dynamics; that is, it exhibits both flows and jumps. The field of hybrid dynamical systems is relatively recent and encompasses a broad range of phenomena and is often used to model natural processes. In this work, we study the maximum number of limit cycles in some classes of discontinuous piecewise hybrid differential systems formed by two rigid systems separated by a straight line. These rigid systems are formed by a linear center plus a homogeneous polynomial of degree 2, 3, 4, 5, and 6. Interest in studying limit cycles of discontinuous piecewise hybrid differential systems has increased in recent years due to their wide range of applications and their connection to the classical *Hilbert's 16th problem*, which asks whether there exists an upper bound for the number of limit cycles in planar polynomial vector fields of a given degree. At present, this problem remains open. Our main objective is to determine the maximum number of limit cycles that can arise in these systems under study; that is, to solve the extended version of *Hilbert's 16th problem* for this class of discontinuous piecewise differential systems. The main result states that the maximum number of limit cycles for this class of systems is at most one when $m = n$, and this bound is achieved. For $m = 2$ and $n = 3, 4, 5, 6$, the maxima are 2, 3, 4, and 5, respectively. For $m = 3$ and $n = 4, 5, 6$, the corresponding bounds are 4, 2, and 2. In the case $m = 4$, the maximum is 6 for $n = 5$ and 10 for $n = 6$. Finally, when $m = 5$ and $n = 6$, the system admits at most 8 limit cycles. In this way, we solve the extended version of *Hilbert's 16th problem* for a class of discontinuous piecewise hybrid differential systems separated by the straight line $y = 0$ and formed by two rigid polynomial differential systems. This is joint work with Jaume Llibre (Autonomous University of Barcelona, Spain) and Durval J. Tonon (Federal University of Goiás, Brazil).

Keywords:

Hybrid systems, limit cycles, Poincaré map.

José Ramón Madrid Padilla

Title

What Is the Smoothest Local Average?

Affiliation:

Virginia Tech, USA

Abstract

Averaging is one of the simplest ways to smooth discrete data, but different averaging kernels can produce very different levels of regularity. This leads to a natural extremal question: among all local averages supported at a fixed scale, which one minimizes the largest possible higher-order oscillation of the resulting signal?

We formulate this problem using convolution operators on the integers and measure smoothness through higher-order finite differences. Fourier analysis transforms the operator-norm minimization problem into an extremal problem for algebraic polynomials on an interval. For first- and second-order differences, the optimal kernels are the familiar uniform and triangular averages. For higher-order differences, however, new phenomena appear: sharp solutions are governed by Chebyshev and Zolotarev polynomials, and in some cases the optimal constants involve elliptic functions.

The talk will describe this connection between optimal discrete smoothing, Fourier multipliers, and classical extremal polynomial theory.

This is joint work with José Gaitán and Carlos Garzón.

Keywords:

Discrete smoothing; convolution operators; finite differences; Fourier analysis; extremal polynomials; Chebyshev polynomials; Zolotarev polynomials.

Emilio Vilches Gutiérrez

Title

A GAMLSS-based approach for modeling fruit weight distributions in ‘Hayward’ kiwifruit orchards

Affiliation:

Universidad de O’Higgins (UOH), Chile

Abstract

We model the temporal evolution of the fruit-weight distribution in ‘Hayward’ kiwifruit grown under Mediterranean conditions in central Chile. Fruit development was monitored over three seasons using repeated non-destructive measurements, from which fruit weight was estimated through an empirical geometric model. The resulting data were modeled as a time-dependent family of probability distributions using Generalized Additive Models for Location, Scale, and Shape (GAMLSS). The Box–Cox Cole–Green original distribution provided stable performance across orchards and seasons, capturing the nonlinear evolution of location, dispersion, and asymmetry, including the increase in variability toward harvest. External validation showed good agreement between observed and predicted distributions and quantiles. This distributional approach provides a dynamic description of fruit growth beyond classical mean-based models and offers a flexible framework for forecasting fruit-size distributions.

Keywords:

Actinidia chinensis var. *deliciosa*; fruit growth modeling; statistical distribution modeling; fruit size distribution; predictive horticulture; Mediterranean production systems.

Rosibel Pacheco Banegas

Title

Highlighting the Bonds of Solidarity Between Honduras and Chile

Affiliation:

President of the Honduran Mathematical Society (ASOMATH), Honduras

Abstract

The relationship between Honduras and Chile has been strengthened over decades through cooperation in education, public health, technical assistance, and academic exchange. One of the earliest milestones was the Chilean Pedagogical Mission of the 1940s, which contributed significantly to the modernization of teacher education and secondary education in Honduras. Subsequently, Chilean specialists collaborated in strategic areas such as public administration, geology, mining, disaster risk management, and higher education. Following the 1973 Chilean coup d'état, a number of Chilean academics found refuge in Honduras, where they contributed to the development of the Universidad Nacional Autónoma de Honduras (UNAH). Their work extended beyond teaching, influencing curriculum design, research methodologies, and the modernization of higher education. Among them, Professor Winston Alarcón played an outstanding role in mathematics education, contributing to both academic development and humanitarian efforts following Hurricane Fifi in 1974, when Chilean professionals participated in emergency housing construction and disaster management initiatives. During the 1970s and 1980s, UNAH was undergoing a period of institutional expansion while facing a shortage of highly qualified professionals in mathematics, economics, social sciences, history, and research methodology. The incorporation of Chilean scholars helped address these needs and introduced new intellectual perspectives that transformed teaching and academic practice. Rather than simply receiving foreign professors, Honduras incorporated new ways of understanding society, development, and scientific inquiry, making Chile an important node within a broader Latin American academic network. A particularly significant example of this collaboration occurred in 1977, when, through agreements among the Organization of American States (OAS), UNAH, and the University of Chile, six Honduran mathematics graduates received scholarships to pursue graduate studies in Mathematical Statistics at the International Center for Statistical Education (CIENES) of the University of Chile. Upon returning to Honduras, these graduates established undergraduate and graduate programs in Mathematical Statistics at UNAH and contributed to national institutions such as the Directorate of Statistics and Census, while maintaining scientific collaboration with Chilean statisticians. This shared history illustrates how academic cooperation between Honduras and Chile has transcended individual exchanges, creating lasting institutional capacities, strengthening higher education, and fostering a common vision for scientific development that continues to inspire new generations of researchers in both countries.

Keywords:

Honduras–Chile cooperation; higher education; academic exchange; mathematical sciences; UNAH; University of Chile; scientific collaboration; history of mathematics.

Gustavo Maldonado

Title

A Proportional Positive-State Model for Post-Irrigation Soil-Moisture Recovery Using Open Smart-Irrigation Data

Affiliation:

Universidad Nacional Autónoma de Honduras (UNAH), Honduras

Abstract

Soil moisture is a positive state variable that is commonly interpreted relative to an agronomic or empirical reference level. This work presents a proportional positive-state model for describing post-irrigation soil-moisture recovery by directly modeling the normalized moisture ratio rather than additive moisture differences. The proposed formulation is based on proportional calculus, where the equilibrium state corresponds to the multiplicative reference value, yielding a first-order proportional relaxation model with an explicit analytical solution.

The methodology was evaluated using an open smart-irrigation dataset collected from a commercial strawberry farm in Paraguay, including soil-moisture measurements, flow-meter records, and electrovalve status. Irrigation events were automatically identified from flow-meter data, recovery periods were extracted, and the proportional model was compared with a classical exponential relaxation model through logarithmic error metrics, chronological hold-out validation, bootstrap analysis, and sensitivity tests.

Results showed that the proportional formulation consistently reduced logarithmic prediction errors across most irrigation events while preserving a simple and interpretable mathematical structure. The proposed model provides a compact event-level description of soil-moisture recovery and offers an alternative framework for modeling positive-state variables when the scientific objective is to describe relative recovery toward a reference condition.

Keywords:

Proportional calculus; Soil moisture; Smart irrigation; Positive-state model; Mathematical modelling; Open data.

José David Zúñiga

Title

Multiyear Validation and Mass-Balance Auditing of Richards Simulations on Uniform and Graded Finite-Volume Meshes Using USDA SCAN Observations

Affiliation:

Universidad Nacional Autónoma de Honduras (UNAH), Honduras

Abstract

The Richards equation is one of the most widely used mathematical models for simulating water movement in variably saturated soils. However, numerical accuracy depends strongly on spatial discretization, temporal integration, and the preservation of water mass balance. This work presents a multiyear validation framework for one-dimensional finite-volume simulations using publicly available USDA SCAN soil-moisture observations.

The methodology combines calibration on a single growing season with independent temporal validation over two additional seasons, allowing numerical discretization errors to be distinguished from structural model uncertainty. Uniform, logarithmic, and algebraic graded meshes are compared through convergence analysis, while an independent mass-balance audit evaluates the consistency of temporal integration under refined predictor–corrector schemes.

Results show that surface-focused graded meshes consistently reduce spatial discretization errors and that this improvement transfers across independent validation periods. Nevertheless, numerical mesh errors remain substantially smaller than model–observation discrepancies, indicating that predictive uncertainty is primarily governed by environmental forcing, model structure, and observational uncertainty rather than grid resolution alone. The proposed framework provides a transparent benchmark for multiyear validation, mesh selection, and water-balance auditing in Richards equation simulations, supporting the development of reproducible computational tools for soil-water modelling.

Keywords:

Richards equation; Finite-volume method; Soil moisture; Numerical validation; Graded mesh; Mass-balance audit; Mathematical modelling.

Carlos Miguel Cruz

Title

Proportional Discrete Fourier Analysis of WaPOR Actual Evapotranspiration over Cropland in Comayagua, Honduras

Affiliation:

Universidad Nacional Autónoma de Honduras (UNAH), Honduras

Abstract

Actual evapotranspiration is a strictly positive environmental variable whose interpretation is frequently based on ratios, percentage changes, and seasonal contrasts. This work presents a proportional discrete Fourier framework for analyzing positive agronomic time series by combining proportional arithmetic with the classical discrete Fourier transform. The proposed methodology interprets the logarithmic representation of the series as a proportional space, allowing seasonal variability to be analyzed relative to the geometric mean rather than through absolute deviations.

The methodology was applied to monthly WaPOR actual evapotranspiration (AETI) data extracted from cropland areas in Comayagua, Honduras, using publicly available datasets from FAO WaPOR, FAO GAUL, and ESA WorldCover. The proportional spectrum was compared with the conventional Fourier spectrum to identify dominant seasonal components and evaluate relative versus absolute variability.

Results showed that both approaches identified a dominant annual cycle, while the proportional formulation provided a unit-invariant interpretation of seasonal variability that naturally emphasizes relative changes around the geometric baseline. Rather than replacing classical Fourier analysis, the proposed framework extends its interpretation for strictly positive environmental variables, offering a useful mathematical tool for remote sensing, digital agriculture, and environmental monitoring.

Keywords:

Proportional arithmetic; Discrete Fourier transform; WaPOR; Evapotranspiration; Remote sensing; Mathematical modelling; Precision agriculture.

Marlon M. López-Flores

Title

A Nonlocal Spatio–Temporal Framework for Conidial Dispersal and Infection Risk in Apple Orchards

Affiliation:

Laboratory of Artificial Intelligence, Robotics and Cybernetics (LIARC), Instituto Militar de Engenharia, Rio de Janeiro, Brazil

Abstract

Fungal trunk diseases are an increasing threat to the sustainability and productivity of fruit orchards. In the Maule Region of Chile, fungi belonging to the family *Botryosphaeriaceae* can infect pruning wounds in apple trees, particularly under humid and cool climatic conditions. Their conidia are dispersed through mechanisms operating at different spatial scales, including local transport by rain splash and longer-range transport by wind.

This work proposes a nonlocal spatio–temporal modelling framework that connects healthy tissue, infected tissue, and environmental conidial density. The model incorporates local infection dynamics, nonlocal dispersal through a spatial kernel, and a memory-dependent sporulation process influenced by climatic conditions and infection age. An explicit numerical scheme is used to simulate the coupled evolution of these variables over space and time.

The proposed framework provides a conceptual and computational basis for identifying critical infection periods, evaluating climatic and dispersal scenarios, and supporting preventive management strategies. Its integration with meteorological observations and field data may contribute to the development of digital agriculture tools for fungal disease-risk assessment in fruit orchards.

This work was developed jointly with William Campillay-Llanos, Samuel Ortega-Farías, Gonzalo A. Díaz, and Shehzad Iqbal.

Keywords:

Nonlocal dispersal; fungal trunk diseases; *Botryosphaeriaceae*; conidial dynamics; mathematical modelling; apple orchards; digital agriculture.

Samuel Ortega-Farías

Title

More than 25 Years of Biomathematical Modelling: Advances and Perspectives from the CITRA Experience

Affiliation:

Centro de Investigación y Transferencia en Riego y Agroclimatología (CITRA)–UTALCA, Chile

Abstract

For more than 25 years, the Center for Research and Technology Transfer in Irrigation and Agroclimatology (CITRA) at the University of Talca has promoted the development and application of biomathematical models to address key challenges in irrigated agriculture. Its work originated with the Integral Agroclimatology and Irrigation Service (SIAR), supported by Fondef, and subsequently evolved into a multidisciplinary center connecting scientific research, technological innovation, education, and knowledge transfer.

This presentation provides an overview of CITRA's experience in irrigation management, agroclimatology, precision agriculture, and adaptation to water scarcity. Particular attention is given to the use of mathematical models, agrometeorological information, remote sensing, and field observations to estimate crop water requirements, support irrigation decisions, and improve water productivity, expressed as kilograms of agricultural product per cubic meter of water applied.

The main scientific and technological advances achieved through collaboration with public institutions, private organizations, agricultural producers, and academic partners are discussed. These experiences demonstrate the importance of integrating modelling, monitoring technologies, and knowledge transfer to improve crop productivity and product quality while reducing production costs and environmental impacts. Finally, future perspectives are presented for developing climate-resilient agriculture through digital technologies, artificial intelligence, and integrated biomathematical models capable of supporting decision-making under increasing water scarcity and climate variability.

Keywords:

Biomathematical modelling; irrigation management; agroclimatology; precision agriculture; water productivity; climate resilience.

Rafael A. López-Olivari

Title

Background and Experiences with Biomathematical Tools in Irrigation Science in La Araucanía Region

Affiliation:

Instituto de Investigaciones Agropecuarias, INIA Carillanca, km 10 camino Cajón-Vilcún s/n, Casilla 929, Vilcún 4880000, Chile

Abstract

Mathematical modelling has become a fundamental component of irrigation science by providing quantitative frameworks to describe the complex interactions among climate, soil, plants, and water. In the La Araucanía Region of southern Chile, bio-mathematical approaches have progressively evolved from research applications to practical tools supporting agricultural water management under increasingly variable climatic conditions.

This presentation reviews research experiences focused on the development, calibration, and application of bio-mathematical tools in irrigation science. Particular emphasis is placed on process-based evapotranspiration models, soil water balance modelling, crop phenology simulation, calibration and validation of soil moisture sensors, and the integration of plant water status indicators into irrigation assessment. These approaches combine mathematical modelling with field experimentation, micrometeorological observations, and continuous environmental monitoring to improve the representation of water dynamics in agricultural systems.

Beyond their agronomic applications, these studies illustrate how mathematical models facilitate the interpretation of complex datasets, parameter estimation, uncertainty analysis, and the generation of predictive tools adapted to local environmental conditions. The experience gained in La Araucanía demonstrates the value of interdisciplinary collaboration between mathematics, environmental sciences, and agricultural engineering, where modelling serves not only as a means of describing biological processes but also as a platform for developing decision-support systems for sustainable water management. Future challenges include the integration of mechanistic models with artificial intelligence, remote sensing, and digital agriculture to enhance model robustness and scalability across diverse production systems.

Keywords:

Biomathematical modelling; irrigation water management; phenological stages; climate variability; evapotranspiration modelling; volumetric soil moisture.

William Campillay-Llanos

Title

From Classical Phenological Models to Reaction–Diffusion Equations: A Spatial Bioclimatic Framework for Cherry Phenology

Affiliation:

Núcleo de Investigación en Producción Alimentaria (NIPA), Facultad de Recursos Naturales, Universidad Católica de Temuco, Chile

Abstract

Classical phenological models based on growing degree days have been widely used to predict sweet cherry development. However, these approaches are generally formulated at individual locations, neglecting the spatial variability generated by topography, microclimate, soil heterogeneity, and orchard management.

This work proposes a mathematical framework that extends classical phenological models from ordinary differential equations to reaction–diffusion partial differential equations by interpreting phenology as a continuous spatio-temporal field. The reaction term represents local phenological development driven by thermal accumulation, while the diffusion term captures the effective spatial continuity of environmental conditions and phenological progression throughout the orchard.

The model incorporates spatially distributed agroclimatic information and is solved numerically using finite-difference methods implemented in **MATLAB**, allowing the simulation of current and future climate scenarios. Compared with conventional point-based or interpolation approaches, the proposed framework provides a mechanistic description that simultaneously integrates spatial and temporal dynamics while preserving the biological interpretation of phenological development.

This work establishes a mathematical basis for digital phenology and precision horticulture and provides a flexible framework for future coupling with epidemiological, hydrological, and crop growth models. The study is the result of a collaborative effort combining mathematical modelling and agroclimatology, carried out jointly with Samuel Ortega-Farías and Marlon M. López-Flores.

Keywords:

Reaction–diffusion equations; Cherry phenology; Mathematical modelling; Digital phenology; Partial differential equations; Precision agriculture; Climate change.

Géza Meszéna

Title

Origin and maintenance of biological diversity

Affiliation:

Department of Biological Physics, Eötvös University & Institute of Evolution, HUN-REN Centre for Ecological Research, Budapest, Hungary

Abstract

Biomathematics is a wide field of science, usually interpreted as a collection of independently developed mathematical models of distinct biological phenomena. In particular, the impossibility of a general theory is the prevailing opinion in ecology. My main interest is to develop high level mathematical theory expressing the necessary commonalities of all ecological models.

Propensity for exponential growth is a consequence of the multiplicative nature of population growth. Natural selection is a generic consequence. As unbounded exponential growth is impossible, it is sure that population growth feeds back to the environment affecting the individuals' life history. Dimensionality of this feedback loop, i.e., the number of 'regulating' variables in the loop, majorates the number of coexisting populations. Competitive exclusion operates when the feedback is one dimensional, e.g., when the populations compete for the same food. The ecological niche of a population can be characterized by their impact and sensitivity vectors. Respectively, they describe the population's differential impact on, and differential sensitivities towards, the vector of regulating variables. Structurally stable coexistence of a community of populations requires independent regulations, i.e., linear independence of both the impacts and the sensitivity vectors of the populations. Robustness of coexistence scales with the strength of these linear independences, which are measured by the norm of the wedge product of the vectors. This general formulation can be properly applied even to spatiotemporally variable environments.

This is the general mathematical underpinning of the ecological expectation that coexistence of populations requires sufficient niche differentiation. It is proven that coexistence of a continuum of populations is always structurally unstable, which is the underlying reason for the observed discreteness of species.

The evolutionary consequence of the regulating feedback loop is a fitness function, which itself is changing via the evolutionary process. As adaptive dynamics theory establishes, the evolving fitness function leads to branching evolution, i.e., emergence of new species. This ecology-driven picture of speciation has earned significant empirical support during the last quarter of century.

Last, but not least, with two colleagues of mine, György Barabás and Chris Terry, we organize a biweekly online seminar series on 'Theoretical ecology' since September 2020 in the framework of the International Initiative for Theoretical Ecology. Everybody is welcome to attend: <http://iite.info/seminar> !

Keywords:

Coexistence, limiting similarity, ecological niche, adaptive dynamics, ecological speciation

Braulio Soto

Title

Presentation of the Chile–Honduras Special Issue: *Mathematical and Computational Modelling of Agricultural and Natural Resource Systems*

Affiliation:

Editor-in-Chief, Journal of Natural Resources and Farming (JONRAF)
Universidad Católica de Temuco (UCT), Chile

Abstract

The *Journal of Natural Resources and Farming (JONRAF)* is an international, peer-reviewed, Diamond Open Access journal published by the Universidad Católica de Temuco. The journal disseminates high-quality research in agricultural sciences, natural resources, environmental sciences, forestry, ecology, and related interdisciplinary fields, promoting scientific contributions that support sustainable development and innovation.

As part of the *Chile–Honduras Meeting on Differential Equations in Biological Sciences and Engineering*, JONRAF is pleased to present the Special Issue entitled *Mathematical and Computational Modelling of Agricultural and Natural Resource Systems*. This Special Issue aims to bring together original research that advances mathematical, computational, and data-driven approaches to biological, agricultural, and environmental systems. Contributions involving differential equations, numerical methods, artificial intelligence, optimization, stochastic modeling, remote sensing, and digital agriculture are particularly encouraged.

Mathematical modeling plays an increasingly important role in modern science by providing complementary perspectives to experimental and observational research. Mathematical models make it possible to formulate hypotheses, identify underlying mechanisms, quantify uncertainty, predict system behavior under alternative scenarios, and integrate knowledge across disciplines. As biological and agricultural systems become increasingly complex, mathematical and computational approaches offer rigorous tools that complement empirical studies and support evidence-based decision-making in research, technology, and sustainable resource management.

For additional information about the journal, its editorial scope, and submission guidelines, please visit the official website:

<https://jonraf.uct.cl/index.php/jonraf/en>

Keywords:

Special Issue; Journal of Natural Resources and Farming; mathematical modeling; computational modeling; differential equations; agricultural sciences; natural resources; artificial intelligence; sustainable agriculture.

Epilogue

The *International Meeting on Differential Equations in Biological Sciences and Engineering* represents far more than a scientific event. It is the beginning of a long-term initiative aimed at building an international community where mathematics serves as a common language for understanding and solving complex real-world problems.

This first edition has brought together researchers, students, and institutions from Chile, Honduras, Brazil, Mexico, Spain, Hungary, and other countries, creating new opportunities for collaboration across diverse areas of mathematics and its applications. By connecting expertise in differential equations, mathematical modeling, numerical analysis, stochastic processes, artificial intelligence, computational mathematics, and interdisciplinary applications in biology, agriculture, environmental sciences, and engineering, the meeting seeks to strengthen scientific cooperation while inspiring new generations of researchers.

Beyond the lectures and discussions, the true legacy of this meeting lies in the collaborations that will emerge from it: joint research projects, student exchanges, scientific publications, graduate supervision, and international networks capable of addressing today's global challenges through rigorous mathematical thinking. We hope that the ideas shared during these sessions will continue to evolve into long-lasting partnerships and innovative research with meaningful scientific and societal impact.

The organizing committee sincerely thanks all invited speakers, participants, collaborating institutions, scientific societies, reviewers, and sponsors whose enthusiasm and commitment made this inaugural edition possible. Their support has laid the foundation for what we hope will become a distinguished international forum for mathematical sciences.

Finally, we warmly invite the mathematical community to remain connected and to stay tuned for the next edition of this meeting. We look forward to welcoming both returning participants and new colleagues as we continue expanding this international network, fostering interdisciplinary collaboration, and advancing the role of mathematical modeling in science, engineering, and society.